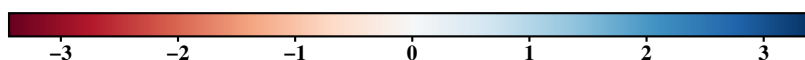


MAE(simple) – MAE(candidate): positive = candidate better

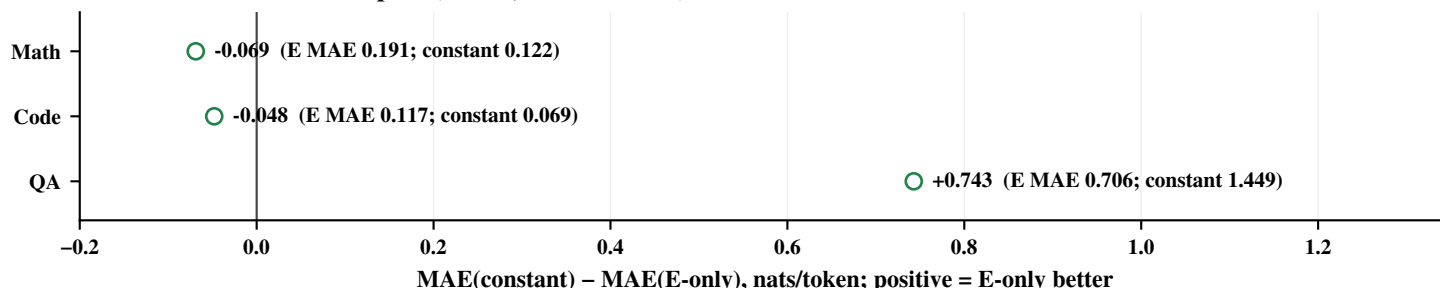


Protocol A

Protocol B

1B@32k	-0.050 MAE 0.254 median	-1.635 MAE 2.272 strength	-0.072 MAE 0.103 bit median	+0.388 MAE 1.050 bit median	-0.025 MAE 0.249 median	-0.241 MAE 1.772 strength	-0.002 MAE 0.028 bit median	+0.021 MAE 0.606 bit mean
1B@96k*	+0.071 MAE 0.193 median	-1.029 MAE 1.302 strength	+0.037 MAE 0.074 bit median	+0.223 MAE 0.397 bit median	not evaluated	not evaluated	not evaluated	not evaluated
1B@112k	-0.018 MAE 0.191 median	-1.087 MAE 1.398 strength	-0.035 MAE 0.071 bit median	+0.125 MAE 0.676 bit mean	-0.028 MAE 0.223 median	+0.377 MAE 0.957 strength	-0.000 MAE 0.038 bit mean	+1.986 MAE 1.120 bit mean
6.9B@32k	-0.316 MAE 0.476 zero	-1.360 MAE 1.635 median	-0.220 MAE 0.253 zero	-2.138 MAE 3.972 bit median	-0.024 MAE 0.160 median	-0.557 MAE 0.705 median	-0.008 MAE 0.032 bit median	-0.315 MAE 0.489 bit mean
6.9B@112k	-0.344 MAE 0.436 median	-1.931 MAE 2.121 strength	-0.229 MAE 0.271 bit median	-3.443 MAE 3.873 bit mean	-0.035 MAE 0.131 median	-0.146 MAE 1.441 strength	-0.006 MAE 0.048 bit mean	+2.477 MAE 1.294 bit mean
	Prune interp d=.9-.6	Prune extrap d=.55	Quantization bits ≥ 4	Quantization int3	Prune interp d=.9-.6	Prune extrap d=.55	Quantization bits ≥ 4	Quantization int3

Distillation · unseen U225 pool (6 runs, Gemma-3-1B)



*1B@96k: only d=.65/.55 and int4/int3; original v46 freeze. Cell lines: gain / candidate MAE / selected simple baseline.

Pythia cells average math, code and QA. All values are nats per native token; the distillation panel is separate.